User Manual

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DOCUMENT HISTORY

Version	Date	Changes made	Sign	Reviewer
0.1	2025-12-15	Initial user manual for submission	AF, AG, VG, MM	DN
0.2	2025-12-23	Updated GUI walkthrough, export steps, troubleshooting	AF, AG, VG, MM	DN
1.0	2026-01-12	Final version approved	AF, AG, VG, MM	DN

1 INTRODUCTION

This document is the **User Manual** for the *Advanced Music Transcription System* developed in the course **TSKS23: Project Course in Signal Processing, Communications and Networking (CDIO)** at Linköping University.

The system allows a user to:

- **record** piano audio in real time using a microphone,
- **upload** an audio or MIDI file for offline transcription,
- **transcribe** into structured musical events (note names with timing information),
- **visualize** waveform and spectrogram views,
- **export** results for external use.

This manual is written so that a new user can set up and run the system without assistance from the developers. No prior background in signal processing is required.

2 SYSTEM REQUIREMENTS

2.1 Hardware

- A computer with an available microphone input (for real-time recording).
- A piano/keyboard for testing (the project was validated on a 61-key keyboard in the range C2–C7).
- Optional: MIDI files for reference-based validation.

2.2 Software

- Operating system: Windows or Linux.
- Python: version 3.10 or newer (recommended).
- Required Python packages are listed in `requirements.txt`.

2.3 Audio permissions

For real-time recording, the operating system must allow microphone access for Python. If recording does not work, see Section 8.

3 INSTALLATION AND SETUP

3.1 Installation

1. Download/clone the project repository to a local folder.
2. Open a terminal in the project folder.
3. Install dependencies:

```
pip install -r requirements.txt
```

3.2 Starting the application

Run:

```
python main.py
```

This launches the graphical user interface.

3.3 Actions outside the GUI (important)

Before launching the application:

- Connect and test your microphone in the operating system settings.
- Keep the project folder writable (exports and local database files are saved from the application).

4 SYSTEM OVERVIEW

4.1 What the system does

The system converts piano performance input into a structured transcription. The internal processing includes preprocessing, onset/offset detection, pitch detection, and formatting of results for display/export.

4.2 Main subsystems

The system consists of five conceptual modules:

1. Input Handling System
2. Graphical User Interface (GUI)
3. Backend Processing Engine
4. Database Management System
5. Output and Export Module

4.3 Typical use cases

- **Offline transcription:** upload an audio/MIDI file, transcribe, inspect results, export.
- **Real-time recording:** record audio with a microphone, stop, transcribe, inspect results, export.

5 GRAPHICAL USER INTERFACE

This section describes the user interface and explains what each control does.

5.1 Main layout

The interface is organized into two main regions:

- **Left panel:** file selection, recording controls, transcription settings, results, and export.
- **Right panel:** visualizations in tabs (Waveform with Notes, Spectrogram).

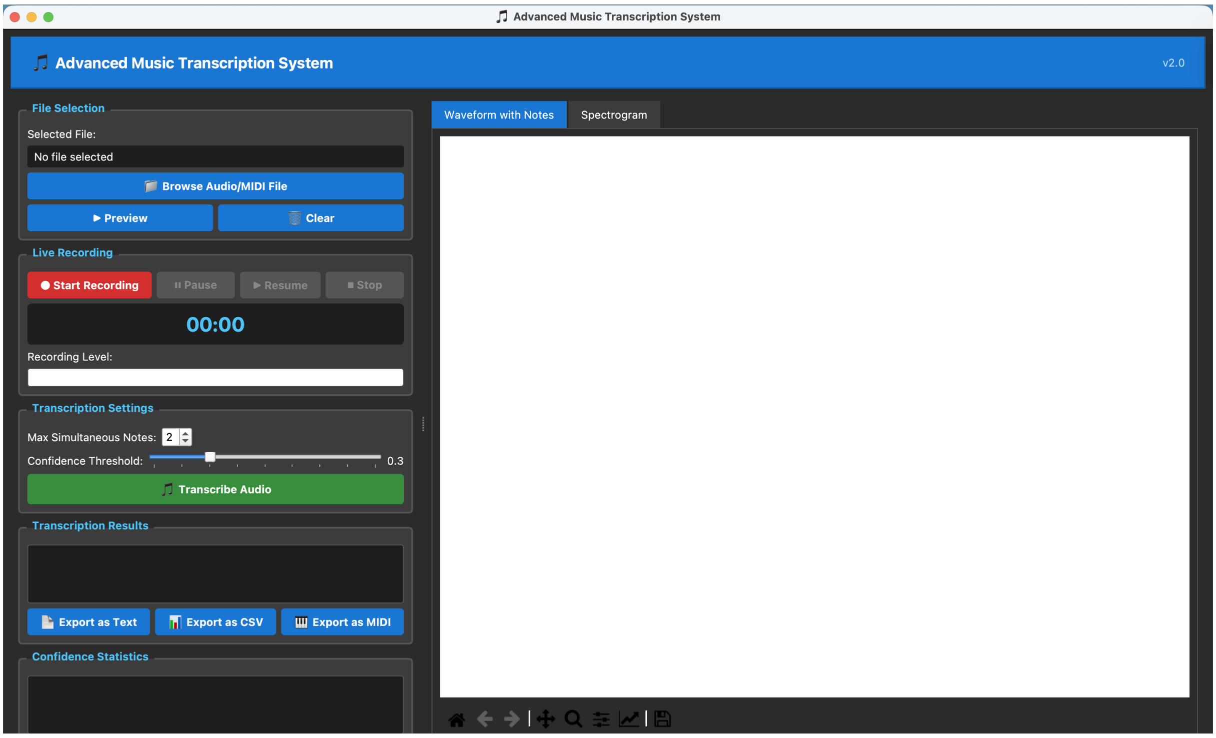

Figure 1: AMT System GUI (example).

5.2 File Selection panel

5.2.1 *Selected File*

Shows the currently selected file. If no file has been chosen, it displays *No file selected*.

5.2.2 *Browse Audio/MIDI File*

Opens a file dialog to select an audio or MIDI file for offline transcription.

5.2.3 *Preview*

Plays back the selected audio file (if supported by the current system setup). Use this to confirm that the correct file was loaded.

5.2.4 *Clear*

Clears the selected file and resets the file-based workflow state.

5.3 Live Recording panel

5.3.1 *Start Recording*

Starts capturing audio from the microphone. The timer begins counting.

5.3.2 *Pause*

Temporarily pauses recording. Use this if you need to stop playing briefly without finishing the session.

5.3.3 *Resume*

Continues a paused recording session.

5.3.4 *Stop*

Ends the recording session. After stopping, the audio becomes available for transcription.

5.3.5 *Recording level*

Displays a real-time input level indicator. If the level is always near zero, check microphone selection/permissions.

5.4 Transcription Settings panel

5.4.1 *Max Simultaneous Notes*

Sets the maximum number of notes to detect at the same time (useful for chords). Start with 1–2 for simple material and increase for chord-heavy recordings.

5.4.2 *Confidence Threshold*

Adjusts a detection threshold used to filter weak detections. If notes are missing, lower it; if too many incorrect notes appear, raise it.

5.4.3 *Transcribe Audio*

Runs the transcription on the currently loaded file or the most recent recording. Progress messages (and errors) appear in the results/log area.

5.5 Transcription Results panel

This area shows the transcription output and system messages. A typical result contains:

- detected note name(s) (e.g. C4, G#5),
- onset time, offset time, and duration,
- optional extra information depending on mode/settings.

5.6 Visualization tabs

5.6.1 *Waveform with Notes*

Displays the waveform of the loaded/recorded audio. Detected note events may be marked on the waveform for visual inspection.

5.6.2 *Spectrogram*

Displays a time–frequency representation of the audio for analysis and debugging.

5.6.3 *Toolbar controls*

The plot toolbar supports common actions such as zoom, pan, reset view, and saving figures.

6 TRANSCRIPTION WORKFLOWS

6.1 Workflow A: File-based transcription (offline)

1. In **File Selection**, click **Browse Audio/MIDI File**.
2. Select a supported file.

3. (Optional) Click **Preview** to confirm the correct audio.
4. Adjust **Max Simultaneous Notes** and **Confidence Threshold** if needed.
5. Click **Transcribe Audio**.
6. Inspect results in **Transcription Results** and in the visualization tabs.

6.2 Workflow B: Real-time recording and transcription

1. In **Live Recording**, click **Start Recording**.
2. Play the piano into the microphone.
3. (Optional) Use **Pause** and **Resume**.
4. Click **Stop** when finished.
5. Adjust settings if needed, then click **Transcribe Audio**.
6. Inspect results and visualizations.

6.3 Recommended first test (quick sanity check)

For a first verification after installation:

1. Record a short clip with **single notes only** (slowly, one note at a time).
2. Use **Max Simultaneous Notes = 1 or 2**.
3. Transcribe and verify note names are plausible and timings are non-zero.

7 OUTPUT, EXPORT, AND SAVED DATA

7.1 What results contain

The transcription output contains (at minimum):

- Note name (e.g. C4, G#5),
- Onset time (s),
- Offset time (s),
- Duration (s).

Results are shown in text form and can be inspected visually using the waveform/spectrogram tabs.

7.2 Exporting results

After transcription, export using the buttons in the results area:

- **Export as Text:** saves a human-readable report.
- **Export as CSV:** saves a table suitable for spreadsheets.
- **Export as MIDI:** saves a MIDI file representing detected notes.

7.3 Saved data / history

The system may store processing outputs locally (e.g. for later retrieval depending on the enabled features). If database/history features are enabled in the current build, transcriptions are saved automatically without requiring manual database interaction.

8 TROUBLESHOOTING

8.1 Recording problems

- **No audio detected / level stays at zero:** check microphone permissions and that the correct input device is available in the OS. Restart the application after changing microphone settings.
- **Very low signal:** move microphone closer to the instrument, increase input gain, and reduce background noise.

8.2 Transcription problems

- **No notes detected:** try lowering the **Confidence Threshold**, and test with clear single notes first.
- **Too many incorrect notes:** raise the **Confidence Threshold**, reduce background noise, and use a smaller max polyphony.
- **Incorrect pitch:** ensure the played notes are within the expected range (C2–C7), and test with a well-tuned instrument.

8.3 Installation errors

- **PyAudio installation errors:** PyAudio can be OS-specific. Confirm that PyAudio and its underlying audio backend are installed correctly for your platform.
- **Missing package errors:** re-run `pip install -r requirements.txt` inside the correct Python environment.

8.4 Export problems

- **Export fails:** choose a folder you have permission to write to (e.g. Documents) and verify disk space.

9 LIMITATIONS AND KNOWN ISSUES

- The system is optimized for piano/keyboard input. Performance may drop for other instruments.
- Background noise and room acoustics can reduce accuracy.
- Dense chords and very fast passages are harder to transcribe reliably than slow, clearly separated notes.
- Depending on the current build and configuration, some features (e.g. history/database browsing) may be partially implemented or limited.